Brianna C. Birkel, Ph.D.

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ORCiD | Google Scholar | GitHub | Linkedin

Summary

Geophysics Ph.D. with experience developing computational methods and open-source Python software for large-scale Earth science data analysis. Research spans seismic imaging, inverse problems, uncertainty quantification, and high-performance computing, complemented by applied machine-learning workflows and evaluation of AI-generated scientific reasoning.

Education

University of Southern California – Ph.D., Geophysics and Seismology Aug 2026

- *Dissertation:* Multiscale Computational Modeling and Seismic Imaging Across Complex Geologic Domains
- *USC Provost Fellow:* Research fellowship for "students whose combination of background and training will make a substantive, documentable, and unique contribution to the program as assessed by faculty." Only one department fellowship awarded per cohort.

University of Washington – B.S., Physics with Interdisciplinary Honors June 2019

Research Experience

Computational Geophysics Ph.D. Researcher, University of Southern California Aug 2019 – Aug 2026

- Developed Python, Bash, and Slurm workflows for 3D spectral-element ground-motion simulations using *Salvus* on high-performance computing clusters, resulting in robust validation of Southern California Community Velocity Models using multi-terabyte sets of observed and synthetic waveform data and a novel goodness-of-fit approach.
- Designed a novel mathematical framework for differential ambient noise tomographic imaging and tested it on synthetic benchmarks to demonstrate reduced model recovery error relative to traditional methods; applied this method as part of a multi-year NSF research project to characterize shallow subsurface structure relevant to fluid flow and landslide processes in the Nepal Himalaya.
- Developed an earthquake catalog locating microseismicity on the Main Himalayan Fault using machine learning–based seismic phase picking, association, and double-difference relocation.
- Collaborated with scientists and engineers across the fields of geology, hydrology, environmental science, climate science, and civil engineering as part of several multidisciplinary research projects; presented computational seismology research at national and international conferences.

Seismology Research Intern, Stanford University May – Sept 2018

- Adapted a complex similarity-search algorithm in Python and Bash to detect brittle-failure upper-mantle earthquakes below Harrat Rahat, Saudi Arabia as part of the Crustal Geophysics Group; presented results at the American Geophysical Union Annual Meeting in 2018.

Quantum Computing Research Assistant, University of Washington, Seattle Jan – Sept 2016

- Characterized nitrogen-vacancy centers in diamond using optical polarization measurements, spectroscopy, and annealing experiments to assess fitness as qubit candidates in the Optical Spintronics and Sensing Lab.

Work Experience

AI Evaluator and Scientific Domain Expert Reviewer, Handshake AI Sept 2026 – Present

- Design and test complex analytical benchmarks from real-world seismological data to assess models' scientific reasoning, developing comprehensive reference solutions and clear scoring rubrics.
- Evaluate AI-generated scientific analyses and presentations for technical accuracy, quantitative reasoning, and interpretation of evidence across physics, geoscience, and environmental science.
- Identify failure modes in AI-generated scientific content, including unsupported conclusions, methodological errors, inconsistent reasoning, and inaccurate interpretation of data and figures.
- Review and provide structured feedback on ambiguous or complex cases for other domain experts, helping improve the reliability and consistency of expert-supervised AI evaluation workflows.

Teaching and Community Engagement

Teaching Assistant, University of Southern California Aug 2019 – May 2026

- Designed and led earth science and physics labs and discussion sections over nine semesters, explaining technical concepts to students from a wide range of majors.
- Earned the highest possible student-review rating from 78% of students in most recent physics course.
- Received average scores of 3.83 out of 4 in student evaluations for "carefully explaining difficult concepts, methods, and subject matter" and 3.73 out of 4 for "[being] receptive to the expression of diverse student viewpoints" in most recent Earth sciences course.

Curriculum Design Volunteer, USC Joint Educational Project October 2021

- Designed age-appropriate experiments for and presented to over 100 elementary school students about earthquake science and emergency preparedness.

Selected Publications and Conference Presentations

Birkel, B. C. (2026). *Multiscale Computational Modeling and Seismic Imaging Across Complex Geologic Domains.* ProQuest Dissertations & Theses Global. https://www.proquest.com/docview/3395749099

Keating, S. A., Clark, M. K., Bickle, M. J., Chen, C.-M., Tipper, E. T., Okaya, D., Ramesh, S., Blunts, P., Atwood, A., Birkel, B. C., Zekkos, D., Chamlagain, D., and West, A. J. (2026). *The role of landsliding in steep mountain hydrology.* Revised manuscript under review at Nature Geoscience.

Birkel, B. C. (2026). *Spatial-VTK: Spatial Validation Tools for Ground-Motion Simulations* (Version 0.1.4) [Computer software]. GitHub. github.com/bcbirkel/spatial-vtk.

Birkel, B. C., & Vidale, J. E. "Comparison of Observed and Simulated Ground Motions in the Los Angeles Basin Against Engineering Baselines." Presented at *Numerical Modeling of Earthquake Motions: Waves and Ruptures*, June 2024, Smolenice, Slovakia.

Birkel, B. C., Vidale, J. E., Juarez, A., & Jordan, T. H. "Comparison of Ground Motions in Earthquakes with SCEC CVM Predictions." Presented at the *Southern California Earthquake Center Annual Meeting*, September 2020, Virtual.

Birkel, B. C., Vidale, J. E., Taborda, R., & Jordan, T. H. "Long-Period Earthquake Ground Motions in the Los Angeles Basin." Presented at *AGU Fall Meeting*, December 2019, San Francisco, CA. Abstract NH23A-08.